

Graph Foundation Models in Science: Navigating Structural Heterogeneity, Geometric Paradigms, and the Path to Autonomous Discovery

The emergence of Graph Foundation Models (GFMs) marks a transformative shift from task-specific Graph Neural Networks to universal architectures capable of cross-domain scientific reasoning. This synthesis examines the critical divergence between text-dependent approaches leveraging Large Language Models and structure-centric frameworks grounded in Riemannian geometry, which argue that intrinsic topology offers superior transferability for text-free scientific data. We analyze recent advancements in defining transferable "graph vocabularies," such as computation trees and spectral prototypes, which enable generalization across heterogeneous feature spaces in biology, chemistry, and physics. While neural scaling laws are increasingly validated in molecular domains, significant contradictions persist regarding their universality in general self-supervised learning, alongside growing concerns over security vulnerabilities like backdoor attacks and model extraction. Furthermore, we highlight the urgent need for standardized benchmarking protocols that account for both topic and format domain shifts to ensure reliable out-of-distribution generalization. Ultimately, the trajectory of GFMs points toward autonomous scientific agents and intelligent laboratories, where geometric rigor and multimodal integration converge to accelerate hypothesis generation and experimental validation, provided that challenges in interpretability, trust, and evaluation standardization are rigorously addressed.

Introduction: The Paradigm Shift Toward Graph Foundation Models in Science

The trajectory of artificial intelligence has been fundamentally altered by the advent of foundation models, which leverage large-scale pretraining to achieve unprecedented generalization across domains. While this paradigm has revolutionized natural language processing and computer vision, its extension to graph-structured data represents a critical frontier for scientific discovery. Graphs serve as the native representation for complex systems in biology, chemistry, and physics, capturing non-Euclidean relationships that traditional sequence-based models cannot inherently process. The transition from task-specific Graph Neural Networks (GNNs) to universal Graph Foundation Models (GFM) marks a significant paradigm shift, aiming to replicate the transferability of Large Language Models (LLMs) for scientific data while addressing the unique challenges of structural heterogeneity and feature alignment [9]. Historically, graph learning has been dominated by models trained from scratch on single datasets, limiting their ability to generalize across the diverse cohorts, assays, and institutions characteristic of scientific research [13]. This limitation is particularly acute in biomedical contexts, where networks vary substantially across studies, necessitating models that can learn transferable structural representations independent of specific node identities or feature schemes [13].

The primary motivation for developing GFMs lies in overcoming the limitations of current GNNs, which often lack the capacity for cross-domain transfer and struggle with out-of-distribution generalization when faced with distribution shifts in graph structure or semantics [108]. Unlike text or images, graphs exhibit extreme heterogeneity in node features, label spaces, and topological structures, making the definition of a shared "vocabulary" for pretraining a non-trivial challenge [9]. Recent efforts have sought to address this by identifying transferable structural units, such as trees, cycles, and computation trees, which can serve as the foundational tokens for a universal graph model [87], [15]. For instance, the concept of a "graph vocabulary" grounded in Riemannian geometry offers a mathematical framework to encode structural invariance, enabling models to learn shared substructures across disparate domains without relying on textual attributes [87], [61]. This structural perspective posits that intrinsic geometry speaks louder than words, providing a pathway to capture complex structural patterns that sequential serialization often fails to preserve [61].

However, the path toward universal GFMs is fraught with contradictions and competing methodologies. A significant divide exists between approaches that leverage LLMs to process text-attributed graphs and those that rely purely on structural learning for text-free graphs. On one hand, integrating LLMs allows for the utilization of rich textual annotations and prior knowledge, with frameworks like UniGraph demonstrating that text can serve as an effective unifying medium for node representations even in domains like molecular graphs [39]. Similarly, LangGFM argues that a large language model alone can be a powerful graph foundation model if effective graph textualization principles are applied [10]. On the other hand, such methods often struggle with numerical features and may fail to capture the full complexity of graph topology due to the limitations of sequential serialization [12]. Conversely, structure-centric models that avoid LLM dependency, such as those utilizing geometric bases or structural prompts, aim to learn transferable representations directly from graph topology, thereby addressing the feature incompatibility that plagues cross-domain applications [274], [94]. These approaches, including GILT, emphasize an LLM-free and tuning-free architecture to handle the extreme heterogeneity of graph data where each graph possesses a unique feature space and topology [121].

In the realm of scientific applications, particularly in biology and chemistry, the need for generalizable models is acute. Molecular graphs, gene regulatory networks, and protein interaction maps vary significantly across studies, yet they share underlying physical and chemical principles that a GFM could potentially capture [13], [116]. Approaches like MolGPS demonstrate that scaling GNNs across vast collections of molecular graphs can yield substantial performance gains in downstream drug discovery tasks, suggesting that the benefits of scale observed in language modeling may indeed translate to graph domains [28]. Similarly, frameworks designed for materials science, such as those utilizing universal directed-tree process graphs, illustrate how multi-modal pretraining on heterogeneous data can accelerate the discovery of new materials by learning semantically rich embeddings that transfer effectively to specialized prediction tasks [75], [51]. Furthermore, recent work on billion-scale models indicates that loss decreases predictably as either model capacity or training data scales, providing the first neural scaling laws for general graphs and reinforcing the viability of large-scale pretraining [14].

The challenge of structural heterogeneity is further compounded by the diversity of task formulations in science, ranging from node-level classification to graph-level property prediction. Traditional GFMs often assume a fixed architectural backbone, which may not be optimal for tasks with varying architectural requirements, necessitating adaptive frameworks that can modify their inference strategies dynamically [70], [24]. Additionally, the integration of domain-specific knowledge, such as molecular motifs or physical constraints, remains essential for achieving high fidelity in scientific predictions, as purely data-driven approaches may overlook critical chemical or physical semantics [205], [289], [275]. As the field matures, the consensus is shifting toward hybrid architectures that combine the semantic understanding of language models with the structural rigor of geometric deep learning, aiming to create robust, scalable, and truly universal models for scientific graph analysis [16], [133], [270]. This synthesis of geometric invariance and semantic flexibility represents the next generation of graph intelligence, moving beyond simple representation learning to solving complex graph-structured applications with autonomous agents [61].

Defining the Scope and Characteristics of Scientific GFMs

The emergence of Graph Foundation Models (GFMs) represents a paradigmatic shift in scientific machine learning, fundamentally redefining the scope of adaptability compared to conventional pre-trained models. While traditional Graph Neural Networks (GNNs) are often constrained to specific datasets or narrow task formulations, scientific GFMs are characterized by their capacity to learn universal representations that transfer effectively across diverse downstream tasks with minimal fine-tuning. This transition is critical in domains such as drug discovery and materials science, where the high cost of experimental validation necessitates models capable of generalizing from sparse, heterogeneous data to complex, multi-step reasoning processes. The defining characteristic of these models is not merely their scale, but their ability to internalize structural and semantic patterns that transcend individual graph topologies, enabling applications ranging from node classification to the modeling of dynamic physical processes [108; 9]. Unlike standard models that require extensive labeled data for each specific task, GFMs leverage the pretrain-transfer paradigm to serve as foundational infrastructure for open-ended reasoning over structured data, bridging the gap between static prediction and the dynamic process modeling essential for modern scientific inquiry [3; 8].

A primary differentiator of scientific GFMs is their sophisticated approach to data heterogeneity and the integration of multi-modal information. Conventional models frequently struggle with the disjointed nature of scientific data, which spans unstructured text, molecular structures, and numeric measurements. In contrast, advanced GFMs employ universal representations that unify these disparate modalities into a single machine-readable format. For instance, recent innovations have introduced directed-tree process-graph representations that consolidate patent literature and experimental data, allowing multi-modal graph neural networks to learn semantically rich embeddings that generalize across chemical domains [75]. This capability extends beyond static property prediction to the modeling of entire experimental workflows, where graph transformer networks have demonstrated superior performance in predicting multi-step outcomes by capturing hidden interactions and sequence-dependent features that linear models miss [25]. Furthermore, the scope of GFMs is significantly expanded by their ability to incorporate external biomedical knowledge, addressing the limitations of purely data-driven approaches. Knowledge-augmented Graph Machine Learning (KaGML) integrates formal scientific laws and informal experimental rules to enhance interpretability and performance in low-data regimes, ensuring that expert knowledge is not substituted but rather augmented by the model [48; 295].

The operational mechanism of scientific GFMs relies on pre-training strategies that diverge sharply from standard supervised learning, though the efficacy of these strategies remains a subject of rigorous debate. Approaches such as GROVER utilize node-, edge-, and graph-level self-supervised tasks to encode rich structural and semantic information from millions of unlabelled molecules, significantly boosting performance on downstream benchmarks [119]. However, the benefits of pre-training are not uniform; ablation studies suggest that self-supervised pre-training on small molecular datasets can yield negligible advantages without careful consideration of data splitting, input features, and architecture, indicating that the "foundation" aspect requires more than just scaling data [40]. To address these nuances, novel frameworks like Joint Multi-domain Pre-training (JMP) simultaneously train on diverse chemical domains, treating each dataset as a unique task to foster cross-domain generalization, thereby achieving state-of-the-art results on a wide array of atomic property prediction tasks [239]. Conversely, other works demonstrate that scaling laws do apply to graph learning, with models like MolGPS showing tremendous benefits from increased depth, width, and dataset diversity, outperforming previous state-of-the-art models on numerous downstream tasks [28]. This tension highlights that the definition of a scientific GFM involves not just the volume of data, but the strategic alignment of pre-training objectives with the complexity of the target scientific domain.

Adaptability in scientific GFMs is further refined through advanced transfer learning and prompt engineering techniques designed to mitigate the gap between pre-training and downstream application. Traditional fine-tuning often suffers from negative transfer when domain shifts are significant. To counter this, researchers have developed prompt learning paradigms that adapt pretrained models to new scenarios without altering core parameters extensively. Methods like UniPrompt and GFMate demonstrate that representation-level prompts and test-time tuning can unleash the capabilities of pretrained models across in-domain and cross-domain scenarios, effectively handling distribution shifts from homophilic to heterophilic graphs [207; 173]. Moreover, meta-learning approaches such as MOCHI align pre-training objectives directly with few-shot inference protocols, resolving structural mismatches in prototype-based

readouts and reducing training time while maintaining competitive performance across node, link, and graph classification tasks [101]. The scope of scientific GFMs also encompasses the critical challenge of out-of-distribution (OOD) generalization, where test data often diverges from training distributions. GFMs are increasingly designed to handle distribution shifts arising from changes in graph structure, domain semantics, or task formulations, employing strategies that support heterogeneous task formulations rather than fixed specifications [108]. This is complemented by efforts to ensure reliability under distributional shift, such as the development of distance-aware GNNs that reduce overconfident mispredictions in cardiotoxicity benchmarks, thereby enhancing the trustworthiness of models used in critical drug discovery pipelines [54]. Ultimately, the defining scope of scientific GFMs lies in their evolving ability to synthesize diverse architectural innovations, data strategies, and adaptation mechanisms into cohesive frameworks capable of accelerating scientific discovery across the full spectrum of graph-structured problems.

The Challenge of Structural Heterogeneity and Feature Alignment

The realization of universal graph intelligence is currently impeded by the absence of a canonical vocabulary and the profound heterogeneity inherent in graph data across different domains. Unlike natural language or computer vision, where tokens and pixels offer a degree of standardization, graph-structured data exhibits drastic variations in node features, label spaces, and topological structures. This lack of alignment creates a critical barrier to direct knowledge transfer, necessitating novel approaches to define and extract patterns that are truly unique to graph structures rather than artifacts of specific datasets. The core difficulty lies in the fact that graphs from disparate domains—ranging from social networks to molecular systems—encode distinct phenomena with incompatible feature dimensions and structural semantics, making the construction of a general-purpose Graph Foundation Model (GFM) a non-trivial endeavor [104].

A primary obstacle in this landscape is structural heterogeneity, where the topology of graphs varies significantly in density, scale, and connectivity patterns. Traditional Graph Neural Networks (GNNs) often struggle to generalize because they are tailored to specific graph distributions, failing to capture abstract structural invariants that persist across domains [33]. To address this, recent research has shifted toward structure-centric approaches that treat graph topology as the primary source of transferable knowledge. For instance, the StructureCentric Graph Foundation Model (SCGFM) proposes modeling graphs as metric measure spaces, utilizing learnable geometric bases to define a shared structural coordinate system. By aligning heterogeneous graphs to these bases via Gromov-Wasserstein distances, SCGFM generates structure-aligned latent representations that accommodate diverse topologies without relying on fixed feature dimensionalities [274]. Similarly, other works argue that graph structures provide more consistent cross-domain information compared to node features and graph labels, suggesting that effective GFMs must prioritize structural generalization over feature unification [33].

Parallel to structural challenges is the issue of feature incompatibility. Node attributes in one domain may be high-dimensional vectors, while in another, they may be categorical or entirely absent. This discrepancy prevents the direct application of models trained on one dataset to another. One promising avenue to resolve this is the use of Text-Attributed Graphs (TAGs), where text serves as a unifying medium. Frameworks like UniGraph leverage Large Language Models (LLMs) to encode diverse node attributes into a unified textual representation space, enabling generalization even for graphs that do not naturally possess textual features, such as molecular structures [39]. This feature-centric perspective is further supported by approaches that treat graph structure as a prior and leverage LLM-unified feature spaces to learn refined interaction patterns [69]. However, reliance on text is not universally applicable, prompting the development of feature-agnostic strategies. Approaches such as GraphFM utilize Perceiver-based encoders with learned latent tokens to compress domain-specific features into a common latent space, effectively decoupling the model from the idiosyncrasies of input feature spaces [1]. Furthermore, some methodologies advocate for the use of intrinsic graph properties, such as degree statistics and centrality measures, as structural prompts to embed diverse graphs into a shared representation space independent of specific node identities or feature schemes [13].

The concept of a "graph vocabulary" has emerged as a central theme in addressing these heterogeneity issues, analogous to the role of tokens in language models. Defining transferable units within graphs remains an open question, with various proposals ranging from computation trees to subgraph motifs. The Graph Foundation model with transferable Tree vocabulary (GFT) rethinks transferable patterns as computation trees derived from the message-passing process, treating these trees as tokens to improve generalization and reduce negative transfer [15]. In contrast, other researchers propose generative graph vocabularies, such as GRAVER, which extract key class-specific subgraph patterns through ego-graph disentanglement to facilitate robust fine-tuning [41]. There is also a theoretical push toward viewing common substructures through the lens of functional behavior and intrinsic geometry, as seen in Riemannian approaches that connect transferable substructures to the geometry of the representation space [153]. Despite these advances, contradictions remain regarding the optimal unit of transfer; while some studies emphasize discrete substructures like trees and cycles [87], others argue for continuous spectral parsing to capture both high-frequency local patterns and low-frequency global signals [240].

Furthermore, the challenge extends to the alignment of tasks across different granularities, including node, edge, and graph-level predictions. The heterogeneity of task formulations complicates the development of a single model capable of handling all scenarios. Task-trees have been proposed as unified learning instances to align these heterogeneous

tasks, theoretically demonstrating that pretraining on diverse task-trees induces transferable knowledge [24]. Additionally, the integration of positional and structural encodings (PSEs) is being investigated as a means to provide universal pretrained representations, though empirical results suggest that optimal performance may still require dataset-specific augmentations [27]. The tension between creating a truly universal model and the need for domain-specific adaptation highlights the complexity of the field, where methods must balance the abstraction of structural principles with the preservation of discriminative details necessary for downstream tasks [108]. Ultimately, overcoming structural heterogeneity and feature alignment requires a paradigm shift from learning on single graphs to learning generalities across vast corpora of diverse graph structures, leveraging both geometric insights and semantic unification strategies [16].

Security and Integrity in Scientific Graph Learning

The deployment of Graph Foundation Models (GFM) in high-stakes scientific environments necessitates a fundamental paradigm shift where security is no longer treated as an auxiliary feature but is recognized as a core requirement for scientific integrity. As GFMs transition from theoretical constructs to operational tools in domains ranging from drug discovery to materials science, they inherit and amplify specific vulnerabilities that threaten the reliability of scientific output. The primary concern lies in the susceptibility of these models to adversarial manipulation, particularly backdoor attacks, which can compromise the validity of downstream applications. Research indicates that launching effective backdoor attacks against GFMs is non-trivial due to challenges in effectiveness, stealthiness, and persistence, yet successful compromises can introduce malicious behaviors that survive fine-tuning [201]. Attackers exploit the lack of knowledge regarding downstream tasks during pre-training by designing label-free trigger association modules and node-adaptive trigger generators, ensuring that the backdoor remains stealthy across diverse node features and persists despite parameter updates in downstream adaptation [201]. This persistence is particularly dangerous in scientific contexts where models are frequently adapted to new datasets, as the embedded backdoors could systematically skew results in critical areas such as biomedical analysis or fraud detection without immediate detection.

Beyond the integrity of model outputs, the intellectual property and proprietary nature of scientific data face significant threats through model extraction attacks. GFMs represent valuable intellectual assets embodying extensive computation and domain expertise, making them prime targets for adversaries seeking to replicate their capabilities without incurring pre-training costs [123]. Recent systematic investigations reveal that GFMs substantially expand the attack surface for model extraction, with attackers able to approximate victim models using only a fraction of the original training time while retaining zero-shot inference capabilities [123]. This vulnerability is exacerbated in scientific domains where training data often includes confidential chemical structures or sensitive medical records. For instance, in drug discovery, publishing neural networks trained on proprietary chemical structures poses severe privacy risks, as membership inference attacks can expose confidential molecular data, particularly for molecules in minority classes which are often the most valuable [298]. The intersection of model extraction and privacy leakage creates a dual threat: the erosion of competitive advantage for research institutions and the potential exposure of sensitive scientific data that was intended to remain secure.

The challenge of maintaining scientific integrity is further complicated by the structural opacity and distributed nature of the foundation model lifecycle, which deviates significantly from traditional data science workflows. The shift from specialized predictive models to generalist foundation models introduces distinct upstream and downstream processes where computational resources and decision-making power are distributed among multiple stakeholders, often hindering transparency and reproducibility [118]. This fragmentation challenges the core principles of "Veridical Data Science," specifically predictability, computability, and stability, as variations in human judgment calls during the opaque upstream pre-training phase can induce unacceptable variability in model outputs [118]. In high-stakes environments like medicine or environmental science, where models inform critical decisions, the inability to rigorously test the "scientific null hypothesis" of a foundation model due to restricted access to internal parameters and training data undermines the trustworthiness of scientific findings [118]. Furthermore, the reliance on black-box models that lack mechanistic interpretability makes it difficult to distinguish between genuine scientific discovery and artifacts introduced by data biases or adversarial perturbations.

Trust in these systems is also fractured by geographic and cultural biases embedded in training data, which can lead to unreliable performance when models are deployed in regions or contexts different from their training distribution [197]. The lack of standardized benchmarks and the rapid proliferation of models with varying degrees of documentation further obscure the evaluation of security and integrity, creating an ecosystem where coordinated defense mechanisms are difficult to implement [85]. While some approaches attempt to integrate physical constraints and symbolic reasoning to enhance reliability, such as constraint-aware neurosymbolic frameworks, the fundamental tension between the flexibility of deep learning and the rigorous demands of scientific validation remains unresolved [190]. Consequently, ensuring the security and integrity of GFMs requires a holistic reimagining of the development lifecycle, incorporating robust defense mechanisms against backdoor and extraction attacks, enforcing strict privacy-preserving

protocols, and establishing transparent governance frameworks that align with the principles of veridical data science to safeguard the credibility of scientific inquiry.

Architectural Innovations: From Message Passing to Geometric Foundations

The evolution of Graph Foundation Model (GFM) backbones represents a paradigmatic shift from traditional Message Passing Neural Networks (MPNNs) toward architectures capable of capturing long-range dependencies, encoding geometric priors, and unifying heterogeneous structural patterns. While MPNNs have historically dominated graph learning, they suffer from fundamental limitations such as over-smoothing, over-squashing, and an inability to model global interactions effectively [168]. To overcome these constraints, recent research has increasingly adopted Transformer-based architectures, which leverage self-attention mechanisms to process graph data as sets or sequences, thereby enabling the modeling of all-pair node influences without the depth-induced degradation characteristic of recursive aggregation [246]. However, the direct application of standard Transformers to graphs requires careful integration of structural inductive biases, leading to diverse architectural innovations ranging from sparse attention mechanisms to geometric deep learning frameworks.

A primary challenge in adapting Transformers to graphs is the incorporation of topological information into the attention mechanism, which is inherently permutation-invariant and structure-agnostic. Early solutions relied on adding positional and structural encodings (PSEs) to node features, yet the generalizability of these encodings across disparate domains remains a critical bottleneck [27]. To address scalability and expressivity simultaneously, novel sparse attention mechanisms have been proposed. For instance, EXPHORMER introduces a framework utilizing virtual global nodes and expander graphs to achieve linear complexity while maintaining theoretical guarantees on spectral expansion, effectively bridging the gap between local message passing and global attention [236]. Similarly, SGFormer challenges the necessity of deep, multi-head attention stacks, demonstrating that a simplified one-layer attention mechanism can achieve competitive performance on billion-scale graphs with significantly reduced computational overhead [221]. These developments suggest that efficient, scalable Transformer backbones are becoming viable alternatives to deep GNNs for industrial-scale applications.

Beyond attention mechanisms, a significant line of inquiry focuses on redefining the input representation of graphs to align with the sequence-processing strengths of Transformers. This "graph-to-sequence" approach treats substructures or traversal paths as tokens, creating a transferable vocabulary analogous to words in natural language. GFT proposes treating computation trees derived from the message-passing process as transferable tokens, thereby reducing negative transfer across domains [15]. Expanding on this, GraphGPT utilizes Eulerian paths to losslessly convert graphs into reversible token sequences, enabling the use of standard large language model backbones for generative pre-training and scaling to billions of parameters [60]. Complementing this, G2PM represents graph instances as sequences of substructures via random walks, employing generative pre-training to learn transferable representations that scale effectively with model size, outperforming generative approaches that plateau at smaller scales [168]. These methods collectively argue that defining a robust structural vocabulary is essential for achieving the cross-domain generalization expected of foundation models.

Parallel to sequence-based approaches, there is a growing consensus that geometric foundations offer a more principled way to encode structural invariance and transferability. This perspective posits that graphs should be viewed as metric measure spaces or manifolds, where transferable knowledge resides in the intrinsic geometry rather than discrete substructures. SCGFM exemplifies this by modeling graphs via learnable geometric bases and aligning heterogeneous topologies using Gromov-Wasserstein distances, effectively creating a shared structural coordinate system [274]. Similarly, the work on Riemannian geometry argues that intrinsic manifold structures provide a superior framework for graph learning compared to extrinsic embeddings, proposing Neural Vector Bundles to parse local coordinates and flatten geometrically compatible spaces for transfer [153]. This geometric turn is further reinforced by RiemannGFM, which constructs product bundles to incorporate diverse geometries of trees and cycles, enabling cross-domain transferability through Riemannian layers [87]. Theoretical analyses support this shift, suggesting that Riemannian geometry captures complex structural patterns inherent to non-Euclidean data more effectively than text-based serializations [61].

Despite these architectural advances, the issue of feature heterogeneity and the disconnect between pre-training objectives and downstream tasks persist. Traditional GFMs often struggle when node feature spaces are incompatible

across domains. To mitigate this, hybrid designs like GraphFM employ Perceiver-based encoders with learned latent tokens to compress domain-specific features into a common latent space, facilitating multi-graph pre-training across diverse datasets [1]. Furthermore, the rigidity of fixed architectural backbones has been identified as a limitation, prompting the development of adaptive frameworks that discover and mix task-specific linear graph operators at inference time to handle heterogeneous architectural requirements [70]. Additionally, meta-learning approaches such as MOCHI align pre-training with few-shot inference protocols by fitting closed-form ridge readouts during training, thereby avoiding the miscalibration issues common in prototype-based readouts [101]. These innovations highlight a trend toward more flexible, adaptive, and geometrically grounded backbones that move beyond the constraints of standard message passing to enable true foundation-scale graph intelligence.

Transformer-Based Architectures and Scalable Attention

The evolution of graph learning from localized message-passing paradigms to global geometric foundations is increasingly defined by the development of transformer-based architectures capable of handling billion-scale graphs. A primary obstacle in this transition has been the quadratic computational complexity inherent in global self-attention mechanisms, which renders standard transformers impractical for large-scale topological data. To address this, recent research has pivoted toward sparse attention mechanisms, novel tokenization strategies, and heterogeneous batching techniques that preserve expressive power while ensuring linear scalability. These innovations are critical for establishing neural scaling laws for arbitrary heterogeneous graphs, moving beyond the limitations of small-molecule or citation network benchmarks [14; 168].

A dominant strategy for mitigating quadratic complexity involves the introduction of sparsity into the attention mechanism, though the philosophical approaches to achieving this vary significantly. Several frameworks propose dynamically sampling relevant nodes or utilizing structural priors to limit the attention scope. For instance, the Deformable Graph Transformer introduces a mechanism that performs sparse attention via dynamically sampled relevant nodes, constructing multiple node sequences based on structural and semantic proximity to achieve linear complexity [269]. Similarly, EXPHORMER leverages virtual global nodes and expander graphs to create a sparse attention mechanism that maintains desirable theoretical properties, such as spectral expansion, while reducing complexity to linear scale [236]. In stark contrast to these structurally complex sparse models, SGFormer challenges the necessity of deep, multi-head architectures entirely, demonstrating that a simplified one-layer attention model can propagate information among arbitrary nodes with linear complexity, achieving significant inference acceleration on web-scale graphs like ogbn-papers100M [221]. Furthermore, LargeGT advances this by combining fast neighborhood sampling with a local attention mechanism that encompasses a 4-hop reception field through efficient 2-hop operations, integrated with a global embedding derived from an approximate codebook [143]. These divergent paths highlight a central tension in the field: whether scalability is best achieved through sophisticated structural sparsification or through radical architectural simplification.

Parallel to sparse attention, the method of converting graph structures into sequences suitable for transformer processing has emerged as a pivotal area of innovation, effectively bypassing the non-Euclidean constraints of traditional graph processing. Traditional approaches often struggle with the non-sequential nature of graphs, but recent works utilize Eulerian paths and substructure sequences to create lossless, reversible mappings. GraphGPT introduces the Graph Eulerian Transformer (GET), which converts graphs into token sequences via (semi-)Eulerian paths, enabling the application of standard transformer backbones to graphs of arbitrary sizes without specialized architectural modifications [60]. This serialization allows for generative pre-training tasks such as next-token prediction, facilitating scaling to billions of parameters. In a similar vein, AUTOGRAPH employs a reversible "flattening" process to transform graphs into random sequences, enabling decoder-only transformers to generate large attributed graphs with linear scaling relative to the number of edges [29]. Additionally, G2PM represents graph instances as sequences of substructures using a random walk-based tokenizer, demonstrating that generative pre-training over these sequences yields representations that continue to improve with model size, unlike traditional message-passing networks that often plateau [168].

The pursuit of true Graph Foundation Models (GFMs) necessitates not only architectural efficiency but also the ability to handle extreme heterogeneity in node features and graph topologies through advanced batching and pre-training strategies. GraphBFF presents a comprehensive framework for building billion-parameter GFMs on arbitrary heterogeneous graphs, introducing specific batching strategies like KL-Batching and Round-Robin Batching to manage industrial-scale data [14]. This work establishes the first neural scaling laws for general graphs, indicating that loss decreases predictably as model capacity and training data scale together. Addressing feature heterogeneity, GraphFM utilizes a Perceiver-based encoder with learned latent tokens to compress domain-specific features into a common latent space, enabling multi-graph pre-training across diverse domains ranging from molecules to product graphs [1]. Meanwhile, OpenGraph tackles the challenge of generalizing to unseen graph properties by employing a unified graph tokenizer and leveraging Large Language Models for data augmentation, achieving remarkable zero-shot performance [63].

Despite these advances, theoretical concerns regarding the expressive power of sparse and higher-order transformers remain central to architectural design. Research indicates that while dense higher-order transformers may be less expressive than the k -Weisfeiler-Lehman test without additional structural information, sparse variants utilizing neighborhood-based attention can achieve expressiveness equivalent to k -WL while significantly reducing computational costs [264]. This suggests that sparsification, when grounded in graph structure, does not necessarily compromise theoretical capacity and may even enhance it by providing explicit structural inductive biases. However, challenges persist in ensuring robust search capabilities and algorithmic reasoning as graph sizes increase, with some studies suggesting that simply scaling parameters may not resolve fundamental limitations in learning search tasks on larger graphs [231]. Consequently, the field is moving toward hybrid designs that balance local structural fidelity with global context, such as NAGphormer, which treats each node as a sequence of tokens aggregated from different hops to learn informative representations in a mini-batch manner [181]. These collective efforts underscore a shift from merely adapting NLP transformers to graphs toward designing native, scalable architectures that respect the unique geometric and combinatorial properties of graph data.

Geometric, Equivariant, and Riemannian Foundations

The inherent non-Euclidean nature of scientific data, ranging from molecular conformations to particle interaction networks, necessitates a departure from standard Euclidean assumptions in deep learning architectures. Traditional Graph Neural Networks (GNNs) often fail to capture the complex geometric priors required for accurate physical modeling, prompting a shift toward frameworks that embed learning within Riemannian manifolds or enforce strict equivariance to physical symmetries. This paradigm shift unifies disparate techniques to better model hierarchical data, molecular dynamics, and particle interactions where flat geometry is insufficient. A foundational argument posits that Riemannian geometry provides a principled basis for graph representation learning, moving beyond isolated techniques to a unifying paradigm that endows networks with intrinsic manifold structures [21]. This perspective is critical because many scientific graphs exhibit hyperbolic or spherical characteristics that Euclidean spaces cannot faithfully represent without significant distortion. Consequently, recent efforts have focused on constructing Graph Foundation Models (GFM) grounded in Riemannian geometry, utilizing structural vocabularies of trees and cycles mapped onto product bundles to achieve cross-domain transferability [87]. Furthermore, the theoretical underpinnings of transferable substructures are increasingly linked to the intrinsic geometry of the representation space, where neural vector bundles enable the parsing of local coordinates to flatten geometrically compatible structures [153].

Parallel to the exploration of intrinsic geometry, enforcing equivariance to physical symmetry groups has become a cornerstone for modeling 3D molecular and particle systems. Standard GNNs often encode 3D geometry using rotationally invariant representations, such as pairwise distances, which inevitably leads to a loss of directional information crucial for predicting properties like potential energy or force vectors. To address this, equivariant GNNs have been developed to operate directly on Cartesian coordinates, incorporating directionality through higher-order representations and novel attention mechanisms that act as spatial-dependent filters [146]. These models are essential for tasks requiring precise 3D reasoning, such as protein-ligand docking and molecular dynamics simulations, where the model must remain invariant to translation and rotation while equivariant to the underlying physical transformations. Advances in pretraining have further leveraged these symmetries; for instance, equivariant energy-based models utilize Riemann-Gaussian distributions to ensure loss functions are $E(3)$ -invariant, enabling robust force prediction from unlabeled 3D molecular data [249]. While equivariant models achieve state-of-the-art accuracy, their reliance on tensor products can be computationally prohibitive. In response, invariant models with linear complexity have been proposed, demonstrating that carefully designed architectures can match the accuracy of equivariant counterparts at a fraction of the computational cost, thus challenging the necessity of high-degree representations for all tasks [247].

The integration of geometric foundations extends to the simulation of large-scale particle systems, where the quadratic scaling of interactions poses significant computational barriers. Hierarchical graph constructions have been introduced to reduce interaction complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N)$, enabling the simulation of gravitational and Coulomb interactions on massive scales while preserving energy conservation and phase space accuracy [261]. Complementing this, physics-aware multiplex graph neural networks separate local and non-local interactions inspired by molecular mechanics, efficiently predicting both scalar and vectorial properties for small molecules and macromolecular complexes [275]. Furthermore, the challenge of learning interaction topologies in lattice systems without predefined links has been addressed by separable graph Hamiltonian networks, which infer underlying particle connections solely from trajectory data while preserving conservation laws [273].

Beyond specific architectural tweaks, there is a growing consensus on the need for universal representations that transcend domain-specific feature spaces. Structure-centric approaches treat graph topology as the primary source of transferable knowledge, aligning heterogeneous graphs to learnable geometric bases via Gromov-Wasserstein distances to create a shared structural coordinate system [274]. This aligns with broader efforts to define foundation models for computational science that fuse modular numerical methods with data-driven learning, ensuring scalability and physics consistency [204]. Additionally, the discovery of governing physical laws from data has been enhanced by combining graph networks with symbolic regression, allowing for the automated rediscovery of Newtonian mechanics and other fundamental laws from observational trajectories [58]. The synergy between geometric deep learning and symbolic discovery is further exemplified by methods that use neural networks to identify self-similarity and scale-transformation symmetries directly from data, bypassing the need for predefined physical models [193]. Collectively, these

developments illustrate a maturing field where geometric rigor and physical symmetry are not merely auxiliary constraints but the foundational bedrock of next-generation scientific machine learning.

Feature-Agnostic Mechanisms and Transferable Vocabularies

The development of Graph Foundation Models (GFM) is fundamentally constrained by the extreme heterogeneity of graph-structured data, where variations in node features, label spaces, and topological distributions across domains prevent the direct application of unified architectures. Unlike text or images, which possess consistent token or pixel spaces, graphs from disparate fields such as social networks, molecular systems, and knowledge graphs often lack compatible input dimensions. To overcome this barrier, recent architectural innovations have pivoted toward feature-agnostic mechanisms that decouple structural learning from domain-specific attributes. This paradigm posits that while raw node features are frequently non-transferable, the underlying computational processes and structural motifs share universal properties that can be encoded into a transferable vocabulary. By treating these structural elements as the primary units of learning, models can achieve cross-task and cross-domain generalization without requiring feature alignment.

A primary instantiation of this approach is the conceptualization of computation trees as the fundamental tokens of a graph vocabulary. In the message-passing process of Graph Neural Networks (GNNs), the aggregation of neighbor information naturally forms tree-like structures rooted at each node. [15] argues that these computation trees, rather than raw features, constitute the transferable patterns shared among different tasks and domains. By rethinking transferable patterns as computation trees, GFT treats these structures as tokens within a unified vocabulary, significantly improving model generalization and reducing the risk of negative transfer. This strategy allows the model to learn generalizable patterns independent of specific feature encodings, facilitating adaptation to diverse graph learning tasks where feature spaces differ drastically. This approach contrasts with methods that attempt to align feature distributions directly, which often fail when the semantic meaning of features shifts between domains.

Complementing the computation tree approach, other frameworks leverage structural invariants and degree statistics to create feature-agnostic representations. [33] emphasizes that graph structures offer more consistent cross-domain information than node features or labels. It introduces a two-phase training process where a structural GFM is first pretrained to predict graph invariants-properties that depend only on the abstract structure, such as centrality measures and community indicators. This structural GFM captures abstract structural information and provides discriminative representations comparable across diverse domains. These representations are then used as positional encodings for a comprehensive GFM that incorporates domain-specific attributes. Similarly, [13] argues that encoding structural prompts based on diffusion-based signatures and degree statistics allows models to embed diverse graphs into a shared representation space without requiring fixed feature dimensions. These methods collectively suggest that structural properties serve as a more stable foundation for pretraining than volatile node attributes.

The theoretical underpinning of these feature-agnostic strategies often relies on the stability and expressiveness of structural encodings. Research into the generalization of positional and structural encodings (PSEs) suggests that while PSEs enhance downstream models, their effectiveness as universal pretrained components depends on their ability to capture intrinsic geometric properties rather than dataset-specific quirks [27]. Furthermore, the concept of structural alignment is formalized in [274], which models graphs as metric measure spaces. This approach employs learnable geometric bases and Gromov-Wasserstein distances to align heterogeneous topologies into a unified coordinate system, effectively creating a structure-aware feature re-encoding mechanism that bypasses the need for compatible input features. This geometric perspective aligns with findings in [153], which explores the intrinsic geometry of representation spaces to parse transferable substructures based on functional behavior rather than discrete motif matching.

Despite the promise of structural vocabularies, challenges remain regarding the adaptability of these mechanisms to varying architectural requirements. [70] highlights that fixed-backbone GFMs often fail when task-dependent architectural attributes differ from those seen during training, suggesting that true generalization requires dynamic adaptation of graph operators at inference time. Additionally, the heterogeneity of tasks-spanning node, edge, and graph levels-necessitates unified learning instances. [24] addresses this by proposing task-trees that align these varying granularities, theoretically proving that pretraining on diverse task-trees induces transferable knowledge capable of efficient adaptation via in-context learning and zero-shot generalization.

The efficacy of these feature-agnostic approaches is further validated by their performance in low-resource and out-of-distribution (OOD) settings. [108] reviews how GFMs aim to learn general-purpose representations through large-scale pretraining to handle distribution shifts arising from changes in graph structure or domain semantics. Methods like [41] enhance robustness by extracting key class-specific subgraph patterns through ego-graph disentanglement and constructing graph vocabularies via graphon-based generative experts. This stabilizes few-shot fine-tuning against structural discrepancies. Moreover, [121] utilizes token-based frameworks that operate on generic numerical features, enabling tuning-free adaptation to new tasks by dynamically understanding class semantics from the context. Collectively, these works illustrate a converging consensus: the path to scalable and generalizable GFMs lies not in forcing feature homogeneity, but in developing sophisticated vocabularies of structural and computational invariants that transcend domain boundaries.

Hybrid Models Integrating LLMs and Tabular Foundations

The evolution of Graph Foundation Models (GFM) is increasingly defined by a departure from rigid, task-specific message-passing architectures toward flexible hybrid systems that leverage the reasoning capabilities of Large Language Models (LLMs) and the in-context learning prowess of Tabular Foundation Models (TFMs). This architectural shift is driven by the need to handle extreme data heterogeneity, where node features and graph topologies vary drastically across domains, a challenge that traditional Graph Neural Networks (GNNs) often fail to address without extensive retraining [9]. A pioneering direction in this space involves repurposing successful tabular models for graph tasks, effectively treating graph nodes as rows in a dynamic table augmented with structural context. The framework G2T-FM demonstrates that TFMs can be transformed into GFMs by augmenting original node features with neighborhood aggregation and structural embeddings before processing them through a TFM backbone [20]. This approach capitalizes on the ability of models like LimiX to handle arbitrary feature types, achieving competitive performance against well-tuned GNNs even in fully in-context regimes. Building on this, GraphPFN advances the paradigm by integrating prior-data fitted networks with attention-based graph neighborhood aggregation, pretrained on synthetic graphs to surpass both task-specific GNNs and earlier hybrid attempts [26]. Conversely, the flow of innovation is bidirectional; BOLERO introduces a lightweight bipartite graph head to augment frozen tabular backbones, providing statistical evidence that explicitly modeling relationships among instances significantly enhances the performance of pretrained tabular transformers [132].

Parallel to the integration of tabular systems, the synergy between LLMs and graph structures has spawned diverse hybrid paradigms aimed at overcoming the limitations of decoupled architectures. Traditional methods often encode textual features independently before aggregating them via GNNs, a process that limits synergistic potential. GraphFormers addresses this by nesting layerwise GNN components directly within transformer blocks, fusing text encoding and graph aggregation into an iterative workflow that captures global semantic perspectives [256]. Similarly, GOFA interleaves randomly initialized GNN layers into a frozen pre-trained LLM, creating a generative model that simultaneously handles semantic and structural modeling without the need for separate alignment stages [133]. For text-attributed graphs, PromptGFM introduces a graph vocabulary learning mechanism where an LLM is explicitly prompted to replicate GNN workflows within the text space, thereby ensuring seamless alignment and transferability across tasks [164]. These methods collectively challenge the necessity of purely structure-based pretraining by demonstrating that language backbones, when properly aligned, can internalize graph topology and semantics effectively. Some approaches even argue that a large language model alone can serve as a powerful GFM through effective graph textualization, repurposing techniques from graph augmentation within the language space [10].

However, the reliance on LLMs is not without contention, particularly regarding their ability to handle numerical features and structural complexity without costly tuning. Critics argue that serializing graphs for LLMs often neglects structural nuances and struggles with non-textual data. In response, frameworks like GILT propose an LLM-free, tuning-free architecture that utilizes a novel token-based framework for in-context learning on generic numerical features, achieving strong few-shot performance without the efficiency bottlenecks of LLM-based baselines [121]. Furthermore, the concept of "graph vocabulary" has emerged as a unifying theme across these hybrid models, suggesting that transferable units-whether derived from substructures, computation trees, or relation tokens-are essential for true generalization. RiemannGFM and Position: Graph Foundation Models are Already Here advocate for structural vocabularies rooted in Riemannian geometry and network analysis, respectively, positing that shared substructures like trees and cycles offer a more robust basis for cross-domain transfer than sequential descriptions [87; 36]. This geometric perspective is further reinforced by SCGFM, which aligns heterogeneous graphs to learnable geometric bases using Gromov-Wasserstein distances, treating topology as the primary source of transferable knowledge [274].

Despite these advancements, significant challenges remain in balancing structural fidelity with the flexibility of foundation models. The debate extends to scalability, where message-passing limitations have prompted the development of generative frameworks like \$G^2PM\$, which tokenizes graph instances into substructure sequences to enable scalable transformer pre-training beyond local neighborhood aggregation [168]. Moreover, the integration of these hybrid models into industrial-scale applications requires addressing data scarcity and distribution shifts, as seen in OpenGraph, which utilizes LLM-enhanced data augmentation and unified graph tokenizers to achieve robust zero-shot

learning [63]. Ultimately, the trajectory of hybrid models suggests a future where the distinction between graph, tabular, and language foundations blurs, driven by architectures that dynamically adapt their inductive biases to the specific structural and semantic demands of the data.

Pretraining Strategies, Scaling Laws, and Learning Paradigms

The evolution of Graph Foundation Models (GFM) marks a decisive shift from training specialized models on isolated datasets to developing generalist systems capable of transferable reasoning across diverse domains. This transition is fundamentally driven by advancements in pretraining strategies that move beyond single-graph self-supervised learning toward scalable multi-graph and multi-task frameworks. Traditional graph self-supervised learning (SSL) often relies on heuristic pretext tasks, such as contrastive learning or generative reconstruction, yet these approaches frequently suffer from homogenization and lack clear generalization capabilities across downstream tasks [11]. To overcome the limitations of dataset-specific training, recent innovations introduce scalable multi-graph pretraining approaches that compress domain-specific features into a common latent space. For instance, methods leveraging Perceiver-based encoders have demonstrated efficacy by training on hundreds of diverse graph datasets, establishing that pretraining on a heterogeneous array of real and synthetic graphs significantly enhances model adaptability and stability [1]. Furthermore, taxonomies categorizing SSL objectives by the scale of knowledge utilized—microscopic, mesoscopic, and macroscopic—provide a structured understanding of how different graph patterns contribute to generalized representations [5].

A critical area of investigation concerns the empirical validity of neural scaling laws within the graph domain. While large language models exhibit predictable performance improvements with increased data and model size, the existence of similar laws for graphs has been contentious. Early benchmarks suggested that many existing graph SSL techniques fail to follow neural scaling behaviors, with downstream performance fluctuating rather than improving consistently as scales increase [7]. However, more recent advancements challenge this pessimistic view by introducing architectures specifically designed for billion-scale operations. The GraphBFF framework presents the first neural scaling laws for arbitrary heterogeneous graphs, demonstrating that loss decreases predictably when both model capacity and training data scale in tandem, provided that architectural bottlenecks are addressed [14]. Similarly, generative pretraining frameworks utilizing substructure sequences have shown continuous performance improvements with model sizes up to 60 million parameters, contrasting with message-passing networks that often plateau or degrade at larger scales [168]. In the molecular domain, extensive studies confirm that GNNs benefit tremendously from increasing depth, width, and dataset diversity, giving rise to foundation models like MolGPS that outperform previous state-of-the-art systems across numerous downstream tasks [28].

The tension between architectural design and scientific utility is further mediated by the choice of learning paradigms, particularly regarding how models adapt to downstream tasks. The traditional "pretrain-then-finetune" paradigm faces challenges related to negative transfer and distribution shifts, prompting the emergence of prompt learning and in-context learning as viable alternatives. Graph Prompt Learning (GPL) aims to bridge the gap between upstream pretraining and downstream scenarios, though consensus on the underlying mechanisms remains elusive. Recent theoretical analyses suggest that representation-level prompts function similarly to fine-tuning classifiers, leading to proposals like UniPrompt that focus on unleashing pretrained capabilities while adapting the classifier to specific scenarios [207]. Alternatively, in-context learning frameworks such as PRODIGY enable models to adapt to novel tasks without parameter optimization by conditioning on prompt examples, offering a tuning-free adaptation mechanism [233]. Despite these advances, contradictions persist regarding the necessity of complex pretraining objectives; some studies indicate that sophisticated masking distributions offer no consistent benefit over uniform sampling, whereas the synergy between prediction targets and encoder architecture proves far more critical [93].

Moreover, the efficiency of pretraining and the handling of data redundancy present significant hurdles for scaling. Empirical evidence suggests that large-scale unlabeled datasets contain substantial redundancy, allowing for coreset construction methods like GraphSculptor to retain nearly full downstream performance while reducing pretraining time by approximately 90% [200]. Concurrently, automated approaches for selecting pretext tasks, such as AUTOSSL and multi-teacher knowledge distillation frameworks, aim to dynamically leverage multiple self-supervised signals to enhance task generalization without manual heuristic design [147; 182]. The integration of structural and semantic diversity through geometric bases or Riemannian geometry offers another pathway to unify heterogeneous graphs, moving beyond feature-based alignment to structure-centric foundation models [274; 87]. Ultimately, the field is converging on the understanding that achieving true foundation model status requires not only scaling data and

parameters but also resolving fundamental conflicts between statistical flexibility and the strict combinatorial constraints inherent to graph structures [8].

The Debate on Neural Scaling Laws in Graphs

The existence and nature of neural scaling laws within the domain of graph self-supervised learning (SSL) remain a subject of intense scrutiny, presenting a stark divergence from the predictable performance improvements observed in Natural Language Processing and Computer Vision. In NLP, the paradigm is characterized by reliable gains as model capacity and dataset size increase, a phenomenon that has driven the development of massive foundation models. However, when attempting to replicate this success in the graph domain, researchers have encountered significant deviations that challenge the direct applicability of these laws. A pivotal study titled [7] challenges the universal presence of these laws, reporting that while SSL loss consistently decreases with scale, downstream task performance often fails to exhibit corresponding improvements. Instead of following a smooth scaling curve, performance tends to fluctuate or plateau, suggesting that mere increases in data volume or model parameters are insufficient without careful consideration of model architecture and pretext task design. This finding implies that the inductive biases required for effective graph learning differ fundamentally from those in sequential or grid-structured data, where scaling has proven more robust.

Contradicting the pessimistic view that graph SSL inherently lacks scaling properties, other large-scale industrial studies present a more optimistic perspective, provided that specific architectural and data engineering constraints are addressed. The paper [14] introduces GraphBFF, a framework that successfully demonstrates neural scaling laws for arbitrary, heterogeneous graphs at an industrial scale. By utilizing a specialized transformer architecture and novel batching strategies, the authors show that loss decreases predictably as either model capacity or training data scales, mirroring the behavior seen in LLMs. This discrepancy between the findings of [7] and [14] highlights a critical nuance: scaling laws in graphs may not be universal across all architectures but are contingent upon the use of models specifically designed to handle graph heterogeneity and scale. The failure to observe scaling in earlier benchmarks may thus be attributed to architectural bottlenecks rather than an intrinsic limitation of graph data itself.

The design of the pretext task emerges as a decisive factor influencing whether scaling behaviors manifest, further complicating the debate. Research indicates that the choice of pretraining objective can dictate the transferability and generalization capabilities of the resulting models. For instance, [199] reveals that many standard graph SSL methods struggle to generalize across datasets, with some performing worse than random initialization when transferred. In contrast, generative approaches like masked autoencoders demonstrate consistent improvements in transfer performance. This suggests that the "scaling" observed is heavily dependent on the semantic richness of the pretext task. Supporting this, [93] finds that sophisticated masking distributions do not necessarily outperform uniform sampling; rather, the synergy between the prediction target and the encoder architecture is paramount. If the pretext task does not align with the structural properties of the graph or the downstream requirements, increasing the scale of pretraining may yield diminishing returns, effectively creating a performance plateau regardless of model size.

Furthermore, the heterogeneity of graph data poses unique challenges to the "one-size-fits-all" scaling hypothesis prevalent in NLP. Graphs vary significantly in topology, feature distributions, and task definitions, which can disrupt the smooth scaling curves seen in more homogeneous domains. [1] addresses this by establishing scaling laws specifically for multi-graph pretraining across diverse domains, showing that training on a wide array of graphs improves adaptability. Similarly, [28] demonstrates that GNNs benefit tremendously from increased depth, width, and dataset diversity in the molecular domain, leading to the creation of MolGPS, a foundation model that outperforms previous states-of-the-art. These works suggest that scaling laws in graphs are likely domain-specific or require a unified approach to graph representation. Without such structural unification, naive scaling of generic architectures may fail to capture the necessary relational inductive biases.

The role of architecture in enabling scaling cannot be overstated, as traditional message-passing networks face limitations such as over-smoothing and over-squashing that hinder performance at scale. [236] and [221] argue that scalable graph transformers with linear complexity are essential for unlocking the benefits of large-scale pretraining. These architectures allow for global context aggregation without the computational prohibitions of full attention, facilitating the training of larger models on massive graphs. Conversely, [40] offers a cautionary note, finding that on small molecular datasets, the benefits of self-supervised pretraining can be negligible compared to supervised methods, and hyperparameter tuning often outweighs the choice of pretraining task. This indicates that scaling laws may only emerge beyond a certain threshold of data and model complexity. Ultimately, the debate centers on whether the lack of

observed scaling in some studies is a fundamental property of graph data or a consequence of suboptimal methodological choices. While [7] posits that current SSL techniques are inadequate for building GFMs due to performance plateaus, works like [14] and [1] provide empirical evidence that scaling is achievable with the right combination of architecture, data diversity, and pretext tasks.

Multi-Graph, Multi-Task, and Synthetic Pretraining

The transition from specialized graph neural networks to generalist Graph Foundation Models (GFM) necessitates a fundamental rethinking of pretraining paradigms to address the extreme heterogeneity of graph data. Unlike text or images, graphs vary significantly not only in their topological structures but also in their node feature spaces, which are often domain-specific and incompatible across datasets. Early investigations into cross-dataset transfer revealed that purely structural pretraining approaches often fail to generalize without sufficient downstream data or feature space alignment, highlighting the difficulty of creating feature-agnostic models [4]. To overcome this, recent frameworks have moved towards compressing diverse domain-specific features into a unified latent space. A pioneering approach in this direction is GraphFM, which introduces a scalable multi-graph pretraining framework utilizing a Perceiver-based encoder with learned latent tokens. By aggregating information from over 150 diverse datasets ranging from molecular structures to citation networks, GraphFM establishes early scaling laws for multi-graph pretraining, demonstrating that exposure to diverse data enhances model adaptability and stability [1]. Similarly, in the chemical domain, Joint Multi-domain Pre-training (JMP) treats datasets from different sources as unique tasks within a supervised multi-task framework, yielding significant improvements in atomic property prediction by leveraging a combined corpus of approximately 120 million systems [239].

Despite the promise of multi-graph corpora, the scarcity of high-quality, large-scale real-world graph data remains a critical bottleneck, prompting a surge in research utilizing synthetic data to instill physical and structural priors. The Prior-Data Fitted Networks (PFN) framework, originally successful in tabular data, has been adapted for graphs through GraphPFN. This model is pretrained on infinite synthetic corpora generated via multi-level stochastic block models and graph-aware causal models, enabling strong in-context learning capabilities and mitigating data scarcity issues that plague traditional GNNs [26]. Extending this concept to scientific discovery, Simulation-Grounded Neural Networks (SGNNs) leverage mechanistic simulators to generate synthetic training data spanning diverse parameter regimes and observational artifacts. This approach allows models to learn latent mechanistic patterns and infer unobservable quantities, such as disease transmission rates, with robustness that exceeds task-specific models trained on limited real-world data [32]. In materials science, the Open MatSci ML Toolkit further explores the utility of synthetic data by employing pretraining tasks based on symmetry operations and synthetic point clouds, revealing that while pretraining may not benefit simple applications, it provides crucial inductive biases for complex, multi-task learning scenarios [23].

The efficacy of these pretraining strategies is closely tied to the ability of models to generalize across distribution shifts and task formulations, yet the literature presents conflicting views on the universal benefits of such approaches. Some studies suggest that self-supervised pretraining on small molecular datasets offers negligible benefits compared to training from scratch, particularly when downstream data is balanced or features are rich [40]. Conversely, other works argue that the lack of benefit stems from insufficient model complexity or inappropriate pretraining objectives rather than a fundamental flaw in the paradigm. To bridge the gap between pretraining objectives and downstream inference, meta-learning approaches like MOCHI align the two by sampling few-shot tasks during pretraining and propagating gradients through a closed-form readout solution. This alignment avoids the structural limitations of prototype-based readouts and significantly reduces training time while maintaining competitive performance across node, link, and graph classification tasks [101]. Furthermore, the integration of physical priors extends to graph network-based simulators, where parameter-efficient conditioning mechanisms allow models trained on specific material types to generalize to unseen constitutive behaviors with minimal fine-tuning data [263].

Scalability remains a central theme, with recent works demonstrating that moving beyond message-passing to generative Transformer architectures can unlock new scaling laws. For instance, representing graphs as sequences of substructures enables generative pretraining that continues to improve performance as model size increases, overcoming the plateau effects observed in traditional contrastive methods [168]. At the industrial scale, the GraphBFF framework provides a recipe for building billion-parameter GFMs on heterogeneous graphs, establishing neural scaling laws that indicate loss decreases predictably as both model capacity and data volume expand [14]. However, the transferability of these models is not guaranteed; success often depends on the alignment between pretraining and downstream feature spaces, and purely structural pretraining may require substantial downstream data to yield improvements [4]. Consequently, the field is increasingly focusing on structure-centric approaches, such as using geometric bases or Riemannian geometry to define a shared structural vocabulary that transcends specific feature

dimensions, thereby facilitating true cross-domain generalization [274]. These diverse strategies collectively highlight that the future of graph foundation models lies in the synergistic combination of multi-graph corpora, synthetic mechanistic supervision, and architectural innovations designed for scalability and structural invariance.

Self-Supervised Objectives: From Contrastive to Generative

The evolution of self-supervised learning (SSL) objectives for graphs represents a fundamental paradigm shift from contrastive discrimination to generative reconstruction and predictive modeling, driven by the unique challenges of non-Euclidean data. Initially, the field was dominated by contrastive approaches that adapted computer vision techniques to graph data, aiming to maximize mutual information between augmented views. These methods typically rely on generating multiple views through topological or feature perturbations and training encoders to discriminate between positive pairs and negative samples [131]. However, the efficacy of these contrastive objectives is heavily contingent on the quality of hand-crafted augmentations and the availability of large batch sizes for negative sampling, which often proves computationally prohibitive [281]. Furthermore, recent investigations have highlighted inherent limitations in contrastive learning, noting that maximizing similarity between perturbed views may fail to capture global graph-level similarities or effectively discriminate between structurally similar graphs that possess distinct properties [155]. These constraints have prompted a critical re-evaluation of contrastive paradigms and accelerated the adoption of generative and predictive alternatives that do not rely on negative sampling.

In response to the limitations of contrastive frameworks, generative objectives, particularly those based on masked autoencoding, have gained significant prominence. Masked Graph Autoencoders (MGAE) operate by reconstructing masked node features or structural elements, drawing inspiration from the success of masked language modeling. A pivotal insight in this domain is that performing reconstruction solely in the raw data space is often unstable due to the non-Euclidean nature of graph data and the high uncertainty involved in inferring masked connections. To mitigate this instability, recent work proposes dual-space reconstruction strategies that operate simultaneously in raw data and high-order latent feature spaces, thereby enhancing the reliability of self-supervision signals [253]. Complementing these architectural innovations, systematic investigations into masking designs have yielded counter-intuitive findings. Rigorous probabilistic frameworks reveal that sophisticated masking distributions often offer no consistent advantage over simple uniform sampling for common node-level tasks. Instead, the choice of prediction target and its synergy with the encoder architecture emerge as the primary drivers of downstream performance, particularly when paired with expressive Graph Transformer encoders [93]. This suggests that the complexity of the pretext task design is often less critical than the semantic richness of the target and the capacity of the backbone.

The transition toward generative objectives is further exemplified by the emergence of large-scale pre-training frameworks that treat graphs as sequences, enabling the application of standard transformer architectures. By converting graphs into token sequences via Eulerian paths, novel approaches facilitate next-token and masked-token prediction, allowing models to scale to billions of parameters [60]. This generative formulation effectively overcomes the scalability bottlenecks typical of traditional message-passing Graph Neural Networks (GNNs). Supporting this trend, research utilizing substructure sequences demonstrates that generative pre-training enables continuous performance improvements with increasing model size, a scaling behavior not observed in many contrastive or reconstruction-based baselines that tend to plateau or degrade [168]. These findings suggest that generative objectives are better suited for the development of universal graph foundation models capable of leveraging massive datasets, as they avoid the expressiveness limits and over-smoothing issues inherent in message-passing mechanisms.

Despite the diversification of objectives, systematic analyses indicate that the architectural backbone and the semantic richness of the prediction target often outweigh the complexity of the pretext task design. Empirical studies of numerous basic SSL tasks suggest that the inherent complexity of graph data requires dedicated pretext tasks rather than direct adaptations from image or text domains [140]. Moreover, in small-data regimes, hyperparameters and data splitting methods can have a larger impact on accuracy than the specific choice of pretraining task [40]. The robustness of different objectives also varies significantly; for instance, in noisy, text-driven biomedical graphs, feature reconstruction proves more robust than relation reconstruction, while contrastive objectives show varying degrees of sensitivity to noise depending on the GNN architecture employed [148].

To reconcile the strengths of different philosophies, recent advancements explore hybrid and multi-task frameworks. These approaches simultaneously optimize multiple pretext tasks, such as mutual information maximization and generative reconstruction, to enhance task generalization across diverse downstream applications [158]. Additionally, theoretical work has begun to provide grounding for predictive models by deriving learning objectives as upper bounds for predicting unobserved latent graphs, addressing the previous lack of theoretical justification for such approaches

[107]. The move toward generative modeling extends beyond node-level tasks to graph generation and optimization, where models utilize random walk-based pretraining to internalize combinatorial rules [8]. Ultimately, while contrastive learning remains a valuable tool, the field is converging on the understanding that generative and predictive objectives, particularly those leveraging transformer architectures and latent space reconstruction, offer a more scalable and robust path toward universal graph foundation models.

Conclusion: Implications for the Future of Scientific Discovery

The trajectory of scientific machine learning is undergoing a profound transformation, shifting from the development of isolated, task-specific Graph Neural Networks to the creation of universal Graph Foundation Models capable of cross-domain generalization. This paradigm shift is driven by the recognition that graphs serve as the native language of complex systems in biology, chemistry, and physics, yet their extreme heterogeneity in topology, feature spaces, and label distributions has historically impeded the transferability seen in natural language processing. The field is currently navigating a critical tension between leveraging the semantic reasoning of Large Language Models and adhering to the rigorous geometric constraints required by non-Euclidean data. While text-attributed approaches offer a unifying medium for multimodal integration, there is a growing consensus that true universality requires structural vocabularies grounded in Riemannian geometry and intrinsic manifold properties. These geometric foundations enable the capture of transferable substructures, such as trees and cycles, without relying on external textual crutches that often fail to preserve the nuanced topological relationships inherent in scientific data [87], [274].

Architectural innovation has moved decisively beyond traditional message-passing mechanisms, which suffer from scalability bottlenecks and over-smoothing, toward transformer-based backbones and generative frameworks that tokenize graph substructures. These advancements have begun to establish neural scaling laws for graphs, demonstrating that performance can improve predictably with increased model capacity and data diversity, particularly in domains with consistent physical laws such as molecular dynamics and materials science [14], [28]. However, the existence of these scaling laws is not universal; it remains contingent on architectural choices that respect graph heterogeneity and pretext tasks that align with downstream objectives. The debate continues regarding the optimal balance between dense global attention and sparse, structure-aware mechanisms, as well as the efficacy of self-supervised objectives in low-data regimes where hyperparameter sensitivity often outweighs pretraining benefits [7], [1].

Despite these technical strides, significant challenges remain in ensuring the reliability, security, and interpretability of these models in high-stakes scientific applications. The opacity of foundation model lifecycles threatens the principles of veridical data science, introducing variability and uncertainty that standard statistical methods fail to capture [118]. Vulnerabilities to adversarial attacks, including backdoors and model extraction, pose severe risks to intellectual property and scientific integrity, particularly in drug discovery and medical diagnostics [123], [201]. Furthermore, the centralization of benchmarking authority risks creating path dependencies that may obscure true model capabilities and limit the diversity of evaluation criteria [85]. Addressing these issues requires a holistic reimagining of the development lifecycle, incorporating robust uncertainty quantification, physics-informed constraints, and decentralized, rigorous evaluation protocols.

Looking forward, the ultimate potential of Graph Foundation Models lies in their evolution from passive predictive tools into active, autonomous agents capable of hypothesizing, designing experiments, and validating discoveries. The convergence of cognitive reasoning with embodied laboratory automation promises to create Intelligent Science Laboratories that can navigate high-dimensional parameter spaces and uncover interdisciplinary connections beyond human cognitive silos [288], [210]. Realizing this vision demands a concerted community effort to curate massive, heterogeneous datasets, standardize evaluation frameworks that account for both topic and format shifts, and develop hybrid architectures that seamlessly integrate geometric deep learning with symbolic reasoning. By bridging the gap between the flexibility of foundation models and the rigidity of scientific validation, the field can unlock a new era of accelerated discovery, transforming graph intelligence into a foundational infrastructure for solving the most complex challenges in science and society.